Task 1 Get Connected

Answer the questions below

Ready? Let's get going!

No answer needed — Correct Answer

Task 2 Understanding SMB

Answer the questions below

What does SMB stand for?

Server Message Block — Correct Answer

What type of protocol is SMB?

response-request — Correct Answer

What protocol suite do clients use to connect to the server?

TCP/IP — Correct Answer

What systems does Samba run on?

Unix — Correct Answer

Task 3 Enumerating SMB

Answer the questions below

Conduct an **nmap** scan of your choosing, How many ports are open?

3 — Correct Answer

What ports is **SMB** running on? Provide the ports in ascending order.

139/445 — Correct Answer

Let's get started with Enum4Linux, conduct a full basic enumeration. For starters, what is the **workgroup** name?

WORKGROUP — Correct Answer

What comes up as the **name** of the machine?

POLOSMB — Correct Answer

What operating system **version** is running?

6.1 — Correct Answer

What share sticks out as something we might want to investigate?

profiles — Correct Answer

Task 4 Exploiting SMB

Answer the questions below

What would be the correct syntax to access an SMB share called "secret" as user "suit" on a machine with the IP 10.10.10.2 on the default port?

smbclient //10.10.10.2/secret -U suit -p 445 — Correct Answer

Great! Now you've got a hang of the syntax, let's have a go at trying to exploit this vulnerability. You have a list of users, the name of the share (smb) and a suspected vulnerability.

No answer needed — Correct Answer

Does the share allow anonymous access? Y/N?

Y ✓ Correct Answer

Great! Have a look around for any interesting documents that could contain valuable information. Who can we assume this profile folder belongs to?

John Cactus ✓ Correct Answer

What service has been configured to allow him to work from home?

ssh ✓ Correct Answer

Okay! Now we know this, what directory on the share should we look in?

.ssh ✓ Correct Answer

This directory contains authentication keys that allow a user to authenticate themselves on, and then access, a server. Which of these keys is most useful to us?

id_rsa ✓ Correct Answer

Download this file to your local machine, and change the permissions to "600" using "chmod 600 [file]".

Now, use the information you have already gathered to work out the username of the account. Then, use the service and key to log-in to the server.

What is the smb.txt flag?

THM{smb_is_fun_eh?} ✓ Correct Answer

Task 5 Understanding Telnet

Answer the questions below

Is Telnet a client-server protocol (Y/N)?

Y ✓ Correct Answer

What has slowly replaced Telnet?

SSH ✓ Correct Answer

How would you connect to a Telnet server with the IP 10.10.10.3 on port 23?

telnet 10.10.10.3 23 ✓ Correct Answer

The lack of what, means that all Telnet communication is in plaintext?

encryption ✓ Correct Answer

Task 6 Enumerating Telnet

Answer the questions below

How many **ports** are open on the target machine?
Note: you may need to scan non-standard ports too.

1 ✓ Correct Answer

What **port** is this?

8012 ✓ Correct Answer

This port is unassigned, but still lists the **protocol** it's using, what protocol is this?

tcp ✓ Correct Answer

Now re-run the **nmap** scan, without the **-p-** tag, how many ports show up as open?

0 ✓ Correct Answer

Here, we see that by assigning telnet to a **non-standard port**, it is not part of the common ports list, or top 1000 ports, that nmap scans. It's important to try every angle when enumerating, as the information you gather here will inform your exploitation stage.

No answer needed — ✓ Correct Answer

Based on the title returned to us, what do we think this port could be **used for**?

a backdoor — ✓ Correct Answer

Who could it belong to? Gathering possible **usernames** is an important step in enumeration.

Skidy — ✓ Correct Answer

Always keep a note of information you find during your enumeration stage, so you can refer back to it when you move on to try exploits.

No answer needed — ✓ Correct Answer

Task 7 Exploiting Telnet

Answer the questions below

Okay, let's try and connect to this telnet port! If you get stuck, have a look at the syntax for connecting outlined above.

No answer needed — ✓ Correct Answer

Great! It's an open telnet connection! What welcome message do we receive?

SKIDY'S BACKDOOR. — ✓ Correct Answer

Let's try executing some commands, do we get a return on any input we enter into the telnet session? (Y/N)

N — ✓ Correct Answer

Hmm... that's strange. Let's check to see if what we're typing is being executed as a system command.

No answer needed — ✓ Correct Answer

This starts a tcpdump listener, specifically listening for ICMP traffic, which pings operate on.

No answer needed — ✓ Correct Answer

Now, use the command **"ping [local THM ip] -c 1"** through the telnet session to see if we're able to execute system commands. Do we receive any pings? Note, you need to preface this with .RUN (Y/N)

Y — ✓ Correct Answer

Great! This means that we are able to execute system commands AND that we are able to reach our local machine. Now let's have some fun!

No answer needed — ✓ Correct Answer

mkfifo — ✓ Correct Answer

Perfect. We're nearly there. Now all we need to do is start a netcat listener on our local machine. We do this using:

"nc -lvnp [listening port]"

What would the command look like for the listening port we selected in our payload?

nc -lvnp 4444 — ✓ Correct Answer

Great! Now that's running, we need to copy and paste our msfvenom payload into the telnet session and run it as a command. Hopefully- this will give us a shell on the target machine!

No answer needed — ✓ Correct Answer

Success! What is the contents of flag.txt?

THM{y0u_g0t_th3_t3ln3t_fl4g} — ✓ Correct Answer

Task 8 Understanding FTP

Answer the questions below

What communications model does FTP use?

client-server ✓ Correct Answer

What's the standard FTP port?

21 ✓ Correct Answer

How many modes of FTP connection are there?

2 ✓ Correct Answer

Task 9 Enumerating FTP

Answer the questions below

Run an **nmap** scan of your choice.

How many **ports** are open on the target machine?

3 ✓ Correct Answer

What **port** is ftp running on?

21 ✓ Correct Answer

What **variant** of FTP is running on it?

vsftpd ✓ Correct Answer

PUBLIC_NOTICE.txt ✓ Correct Answer

What do we think a possible username could be?

mike ✓ Correct Answer

Great! Now we've got details about the FTP server and, crucially, a possible username. Let's see what we can do with that...

No answer needed ✓ Correct Answer

Task 10 Exploiting FTP

Answer the questions below

What is the password for the user "mike"?

password ✓ Correct Answer

Bingo! Now, let's connect to the FTP server as this user using **"ftp [IP]"** and entering the credentials when prompted

No answer needed ✓ Correct Answer

What is ftp.txt?

THM{y0u_g0t_th3_ftp_fl4g} ✓ Correct Answer

Task 11 Expanding Your Knowledge

Answer the questions below

Well done, you did it!

No answer needed ✓ Correct Answer

Certificate